

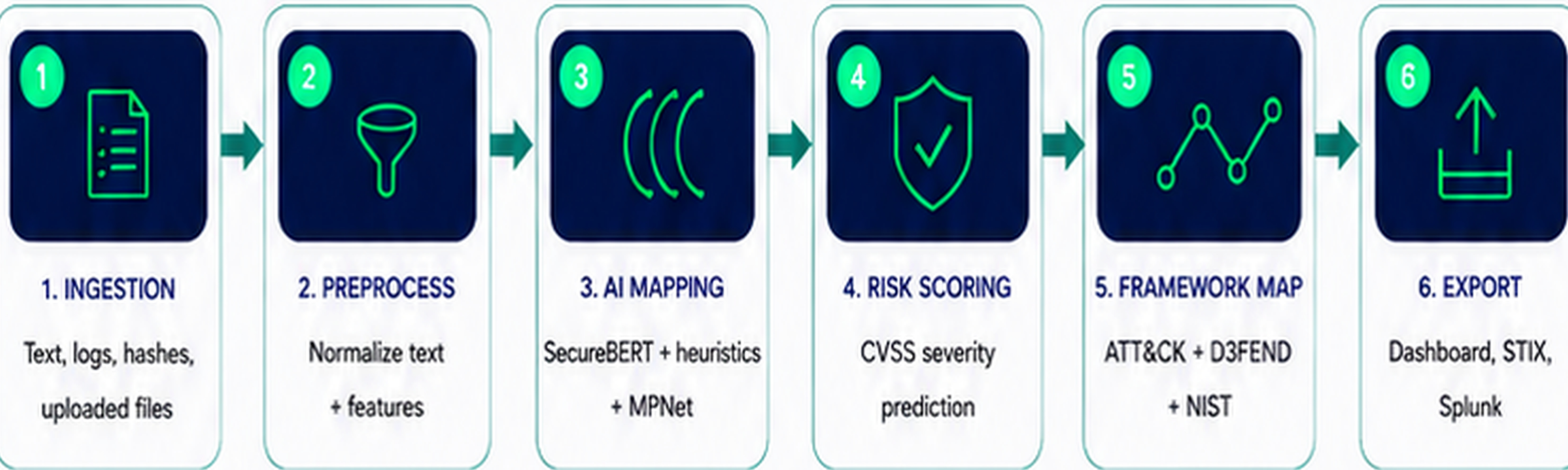
ABSTRACT

autoMITRE automates analysis and framework mapping of unstructured security descriptions, logs, malware hashes and uploaded telemetry. It combines SecureBERT classification, semantic retrieval and dual-LLM mitigation generation to produce MITRE ATT&CK, NIST 800-53, D3FEND and STIX 2.1 outputs for faster SOC decision-making.

TECHNOLOGIES USED



WORK FLOW



RESULTS / EVALUATION

96.65%

SecureBERT F1

98.41%

Framework Mapping

0.8s

Inference Latency

40MB

CSV RAM Footprint

Threat Mapping: SecureBERT + Heuristics + MPNet • Unstructured ATT&CK: RAG + ChronoDB context – 84.06% efficacy

PCAP Analysis: Scapy parsing – 63.20% baseline due to TLS encryption • Report Export: **STIX 2.1** + SIEM exporter

DISCUSSION & CONCLUSIONS

autoMITRE demonstrates the value of hybrid AI in cyber threat intelligence. SecureBERT strengthens classification of cyber language, while RAG constrains generative outputs to validated framework context. The platform converts raw threat data into structured intelligence and actionable guidance.

FUTURE WORK

DBSCAN anomaly discovery; CSA/GDPR matrices; automated SOAR firewall actions.

KEY INNOVATION, LIMITATIONS

INNOVATION: Hybrid deterministic NLP + generative RAG; cross-framework interoperability; dual-LLM routing.

LIMITATIONS: Encrypted TLS traffic constrains PCAP inspection; local inference requires adequate memory.